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Single-Trajectory GRPO for Cooperative Multi-Agent Reinforcement Learning

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Paper + code
github.com/ko120/IGRPO

TL;DR — The n agents inside one episode form the GRPO group: critic-free MARL matches PPO with 16% less peak GPU memory — normalize returns globally over agents and time, not per time step.

1 Why critic-free MARL?

- PPO-style MARL (IPPO, MAPPO) trains a **critic + GAE**: extra memory, extra compute, and value-estimation error.
- **GRPO** drops the critic: advantages are rewards normalized across a *group of rollouts of the same prompt* — the recipe behind LLM post-training (DeepSeekMath).
- In MARL we do **not** get many rollouts from an identical state — so where does the group come from?
- **Our answer:** in homogeneous cooperative MARL, the n agents inside *one* episode already form the group.

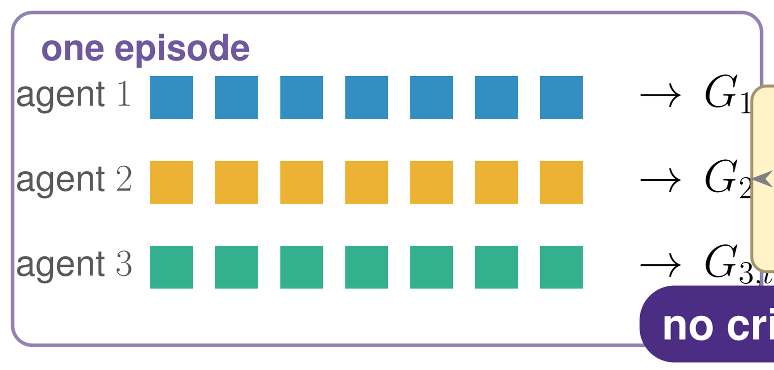
2 Key idea: the agents are the group

Standard GRPO (LLM post-training)

prompt x → $y_1 \to r_1$, $y_2 \to r_2$, $y_G \to r_G$ → group $= G$ rollouts of the *same* prompt: $\hat{A}_k = \frac{r_k - \mu}{\sigma}$

✗ needs many rollouts from the same state — unavailable in MARL

This work: single-trajectory GRPO



group $= n$ agents $\times$ T steps
$$\hat{A}_{i,t} = \frac{G_{i,t} - \mu}{\sigma}$$

no critic • no GAE • no extra rollouts

3 Method

PPO-like clipped objective, KL-regularized, critic-free:

$$J(\theta) = \mathbb{E}\big[\min(r\,\hat{A},\ \mathrm{clip}(r, 1-\epsilon, 1+\epsilon)\,\hat{A})\big] - \beta\,\mathrm{KL}(\pi_\theta \,\|\, \pi_{\mathrm{ref}})$$

value-loss coefficient 0, no entropy bonus, value targets $\equiv 0$. Advantages built from **discounted returns-to-go** $G_{i,t} = \Sigma_\tau \gamma^\tau R(s_{i,t+\tau}, u_{i,t+\tau})$.

A) Per-time-step cross-agent (Zhao & Li, 2025)
$\hat{A}_{i,t} = \frac{G_{i,t} - \mu_t}{\sigma_t}$, (μ_t, σ_t) over the n agents at step t $\Rightarrow$ only $n{=}3$ samples per step, and advantages are **zero-sum across agents at every step**.

B) Global episode statistics (this work)
$\hat{A}_{i,t} = \frac{G_{i,t} - \mu}{\sigma}$, one (μ, σ) over *all* nT values $\{G_{i,t}\}$ $\Rightarrow$ nT-sample baseline that **preserves absolute return differences** within the episode.

Drop-in: swaps RLlib's GAE connector for a group-normalization connector — everything else is standard PPO machinery.

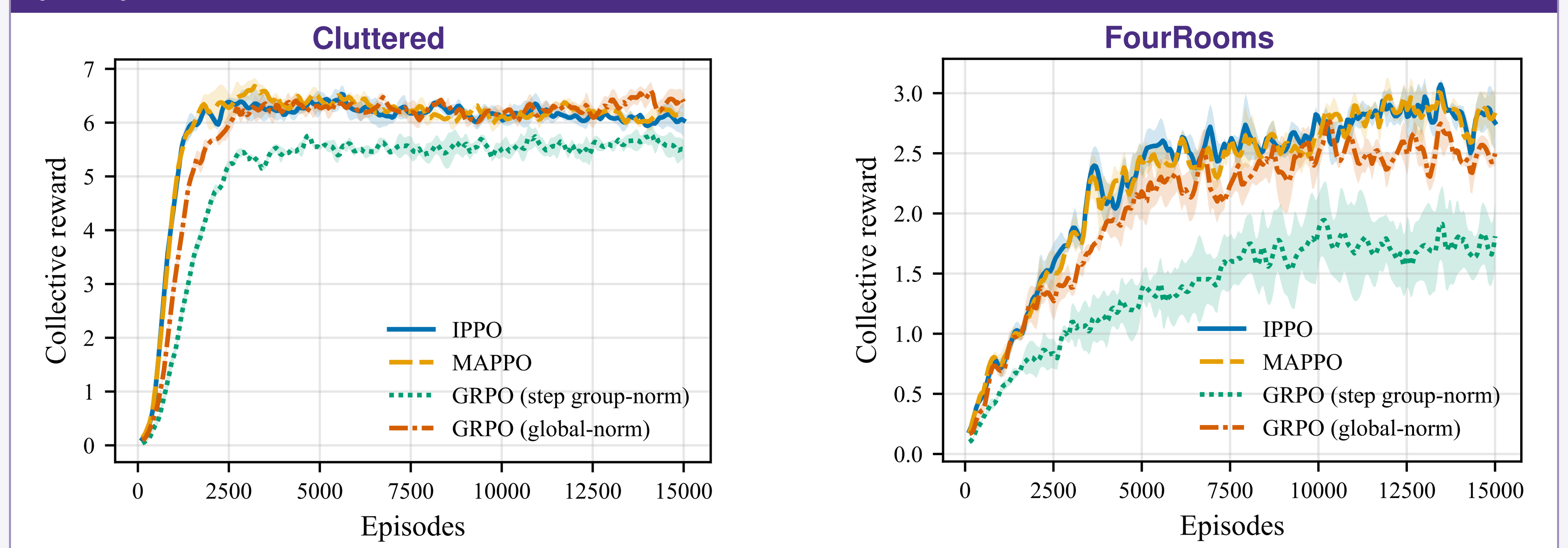
4 Setup: two sparse-reward gridworlds

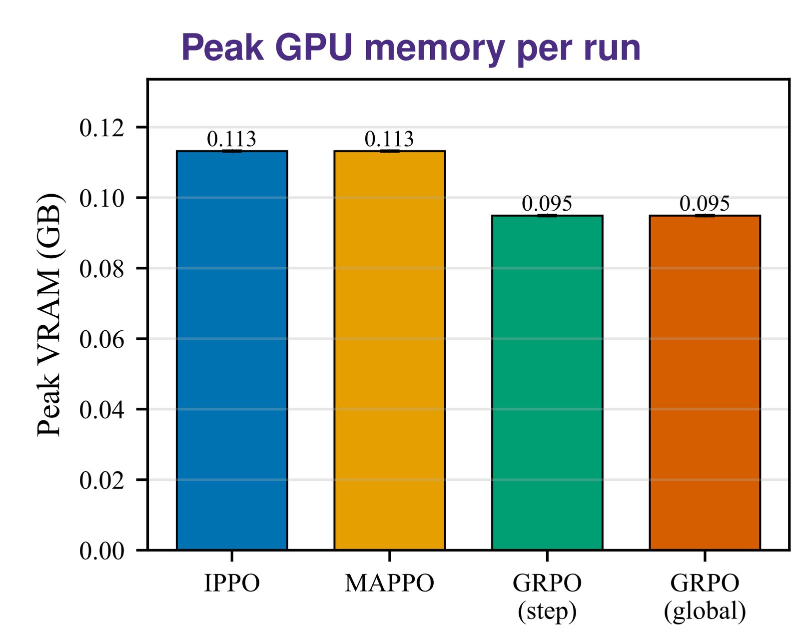
Cluttered 15×15
30 fixed obstacles, fixed goal, randomized starts — moderate difficulty.

FourRooms 15×15
doorways, goal, *and* starts all randomized each reset — hard exploration.

$n{=}3$ agents • egocentric $5{\times}5{\times}3$ obs • 7 actions • ≤ 100 steps • parameter sharing (CTDE) • reward only on reaching the goal: $r = 1 - \frac{\texttt{step\_count}}{\texttt{max\_steps}}$, else 0 • 3 seeds, mean $\pm$ SEM • built on RLlib.

5 Main result — GRPO matches PPO with less memory (RQ3)





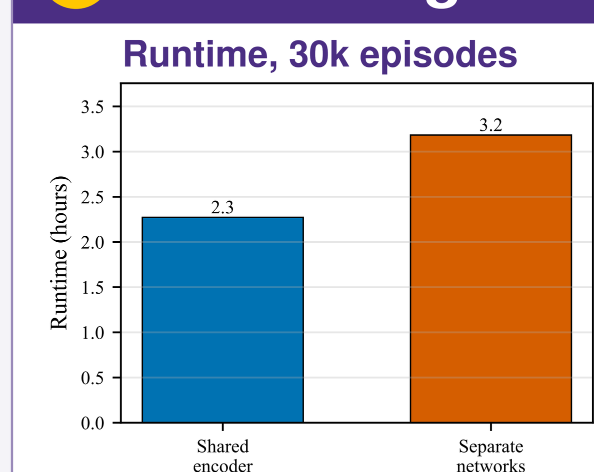
-16% peak VRAM, critic-free
$0.113\,\text{GB} \to 0.095\,\text{GB}$

6.4 ± 0.2 vs. 6.1 ± 0.1
GRPO (global) vs. IPPO — Cluttered

2.5 ± 0.1 vs. 2.8 ± 0.1
GRPO (global) vs. IPPO — FourRooms

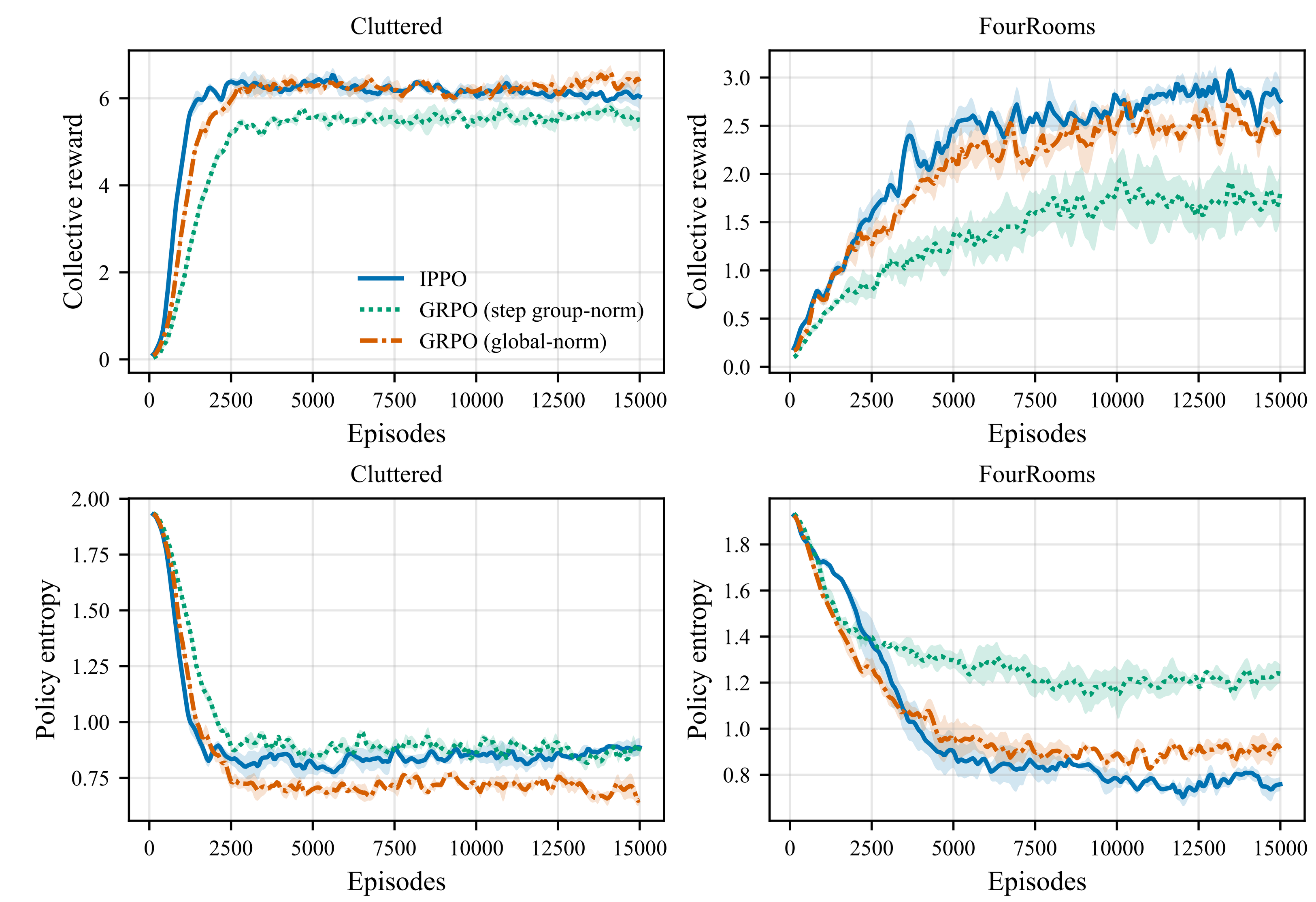
Takeaway: GRPO (global-norm) reaches IPPO-level reward on both environments with **no critic** — comparable wall-clock, 16% lower peak VRAM at this model scale.

6 PPO design choices (RQ1–RQ2)



- **Shared encoder $\approx$ separate networks** in sample efficiency, but **30% less wall-clock** (2.3 vs. 3.2 h).
- **MAPPO = IPPO under sparse rewards**: the centralized critic adds no measurable benefit (6.1 ± 0.1 / 2.8 ± 0.1 for both). More information $\neq$ better.

7 The group statistics are the critical choice (RQ4)



Top: collective reward. Bottom: policy entropy. Cluttered (left), FourRooms (right).

- On Cluttered both variants learn; on FourRooms **per-time-step collapses**: plateaus at 1.7 ± 0.3 (vs. IPPO 2.8 ± 0.1) with entropy stuck high — **global** stays competitive (2.5 ± 0.1).
- **Why:** (μ_t, σ_t) are estimated from only $n{=}3$ returns per step; under sparse reward the three returns are often identical (all zero) $\Rightarrow$ advantage collapses to zero — no learning signal.
- **Zero-sum grading:** even when *all* agents succeed, slower agents get negative advantage.
- Global pools nT samples into one stable baseline; with lateness-scaled rewards bounded in $[0, 1]$, return scales stay comparable across time, so one global mean is a sound baseline.

Takeaways

✓ **Critic-free GRPO is a practical CTDE recipe**: matches IPPO/MAPPO on both tasks with 16% less peak VRAM.

✓ **Normalize globally** over agents $\times$ timesteps — per-time-step cross-agent statistics fail on hard exploration.

✓ **Centralization is not free lunch**: MAPPO's global critic adds nothing under sparse rewards; shared encoders cut runtime 30%.

Limitations: two gridworlds from one family, $n{=}3$, small CNNs (absolute VRAM saving modest); formal variance analysis of single-trajectory baselines left to future work.

References: Schulman et al., PPO, 2017 • Yu et al., MAPPO/IPPO, NeurIPS 2022 • Shao et al., DeepSeekMath (GRPO), 2024 • Zhao & Li, one-step MARL GRPO, 2025 • Liang et al., RLlib, 2018

code & configs: github.com/ko120/IGRPO